The Heisenberg uncertainty principle

Suppose A and B are Hermitian operators, and $|\psi\rangle$ a quantum state. Write

$$\langle\psi|AB|\psi\rangle = x + iy, \quad x, y \in \mathbb{R}.$$

Then

$$\langle\psi|AB|\psi\rangle^* = \langle\psi|(AB)^\dagger|\psi\rangle = \langle\psi|B^\dagger A^\dagger|\psi\rangle = \langle\psi|BA|\psi\rangle\,,$$

thus

$$\langle\psi|\underbrace{[A,B]}_{AB-BA}|\psi\rangle = 2iy \quad \text{and} \quad \langle\psi|\underbrace{\{A,B\}}_{AB+BA}|\psi\rangle = 2x.$$

Taken together,

$$|\langle\psi|[A,B]|\psi\rangle|^2 + |\langle\psi|\{A,B\}|\psi\rangle|^2 = 4\left(x^2+y^2\right) = 4\,|\langle\psi|AB|\psi\rangle|^2. \tag{1}$$

Cauchy-Schwarz inequality applied to $|v\rangle = A\,|\psi\rangle$, $|w\rangle = B\,|\psi\rangle$ $\rightsquigarrow$

$$|\langle\psi|AB|\psi\rangle|^2 \leq \langle\psi|A^2|\psi\rangle\,\langle\psi|B^2|\psi\rangle\,. \tag{2}$$

Thus

$$|\langle\psi|[A,B]|\psi\rangle|^2 \overset{(1)}{\leq} 4\,|\langle\psi|AB|\psi\rangle|^2 \overset{(2)}{\leq} 4\,\langle\psi|A^2|\psi\rangle\,\langle\psi|B^2|\psi\rangle\,.$$

Suppose C and D are two observables. Substituting $A = C - \langle C\rangle$ and $B = D - \langle D\rangle$ leads to the *Heisenberg uncertainty principle*

$$\Delta(C)\cdot\Delta(D) \geq \frac{|\langle\psi|[C,D]|\psi\rangle|}{2}.$$

Interpretation for experiments: repeated preparation of $|\psi\rangle$, measure C in some cases and D in the other cases to obtain standard deviations $\Delta(C)$ and $\Delta(D)$.